



"Anything done manually twice gets automated"

Nikita Petrich

FREELANCE SENIOR FULL-STACK & AI ENGINEER

Phone: +49 15679088678 **Mail:** n.petrich@sequenz.io **Web:** sequenz.io

Location: Remote (preferred) · Munich hybrid: 1–2 days/week · Travel: 1–2 days/month (if needed)

Onboarding: On-site at the start (if needed)

Availability: immediately available · full-time (full capacity) **Rate:** on request

Qualification: IHK-certified Application Developer (Fachinformatiker, Cologne)

Experience: 7+ years

FOCUS AREAS

LLM integration RAG GDPR-Compliant AI Architecture AI-powered automation
 Agentic coding / AI engineering TypeScript Python Backend (NestJS · Node.js · FastAPI)
 Frontend (Next.js · React · Angular) PostgreSQL Clean Architecture Microservices CI/CD
 Docker GDPR-compliant AI architecture

LANGUAGES

German · Native English · B2 · US/EU remote-team experience

WAY OF WORKING

results-oriented self-driven & reliable autonomous, asynchronous way of working
 clear communication (EN/DE)

PROFILES

[Website](#) [LinkedIn](#) [GitHub](#) [freelancermap](#) [Malt](#) [Certificates](#)

i I'm a freelance Senior Full-Stack & AI Engineer focused on AI: **LLM integration, RAG and AI-driven automation** – from architecture all the way to stable production.

My guiding principle: **anything done manually twice gets automated**. That applies to my clients' processes as much as to my own work. In logistics, a route management system I built saves up to 1,000 hours per employee per year through automated document workflows; in retail, **up to 40 hours of admin work per month**; at a notary's office, client phone enquiries dropped by **up to 70%**.

I apply the same logic to development itself. I work consistently with **agentic coding and AI engineering workflows** (Claude Code, Cursor, Code Rabbit, MCP) – spec-driven, with AI-assisted test generation and agentic review processes. This isn't a gimmick, it's the reason for unusual speed at consistent quality: a complete cross-platform app from first commit to handover **in roughly two months**, as the sole developer.

How I work: **independently, remote and asynchronously**. I don't need daily direction, just a clear goal – I'll figure out the path there myself and speak up before anything slips. I document decisions, ship tested and review-ready code, and flag early when a requirement turns out to be technically more expensive than expected. I communicate in German and English.

Technically: **TypeScript and Python**, backends with NestJS, Node.js and FastAPI, frontends with Next.js, React and Angular, plus PostgreSQL, Docker, CI/CD and cloud on Azure or GCP. Clean Architecture and microservices are standard; **GDPR compliance and Secure by Design** are built in from day one.

I've done this inside a **20-person engineering team** – most recently on an AI-powered LegalTech platform with **more than 3,000 customers** – and as a sole developer with full product ownership. Over **seven years** across LegalTech, HealthTech, e-commerce, EdTech and logistics.

Does this fit what you have in mind? Book a [free intro call](#)

Skills & Expertise

Complete overview of my technical skills

01 AI, LLM & AI ENGINEERING

LLM Integration RAG (Retrieval Augmented Generation)
 Prompt Engineering Embeddings Vector Databases
 pgvector Semantic Search Document Analysis OCR
 LangChain OpenAI API Azure OpenAI
 Anthropic Claude API Hugging Face Ollama AI Agents
 Agentic Software Engineering AI-Powered Automation
 AI Engineering Artificial Intelligence
 GDPR-Compliant AI Architecture

03 PROGRAMMING LANGUAGES

TypeScript JavaScript Python C# SQL Bash
 HTML CSS SCSS XML UML

05 FRONTEND

React Next.js Angular React Native Ionic Capacitor
 Cross-Platform Development iOS Android Redux
 NgRx RxJS TanStack Query React Hook Form
 React Router Axios Zod Tailwind CSS shadcn/ui
 Material UI (MUI) Fluent UI Single Page Application (SPA)
 Progressive Web App (PWA) Server-Side Rendering (SSR)
 Internationalization (i18n) Frontend Development
 Web Development

07 DATABASES & DATA

PostgreSQL MySQL MariaDB
 Microsoft SQL Server (MSSQL) MongoDB Mongoose
 NoSQL SQLite Redis Cloud Firestore Prisma
 MikroORM TypeORM Drizzle ORM
 Entity Framework Core Data Modeling Data Architecture
 DTOs Data Synchronization ETL CSV Processing
 Caching Idempotency

09 AUTH & SECURITY

JWT OAuth 2.0 Single Sign-On (SSO) better-auth
 Firebase Authentication (Auth) Firebase Security Rules
 Authentication Role-Based Access Control (RBAC)
 Row Level Security SSL/TLS End-to-End Encryption
 O365 Identity
 Microsoft Entra ID (formerly Azure Active Directory)
 Secure by Design IT Security

02 AGENTIC CODING

Claude Code Cursor AI GitHub Copilot Code Rabbit
 ChatGPT Lovable v0 Claude Code Hooks
 Slash Commands Agent Skills Subagents
 Multi-Agent Workflows Model Context Protocol (MCP)
 Agentic Coding Context Engineering
 Spec-Driven Development Agentic Pull-Request Workflows
 AI-Powered Test Generation

04 BACKEND & FRAMEWORKS

Node.js NestJS Express.js Fastify FastAPI Flask
 ASP.NET Core .NET Core GraphQL REST API OpenAPI
 Swagger WebSockets Webhooks Message Queues
 BullMQ RabbitMQ Headless CMS Strapi Directus
 Backend Development Object-Oriented Programming (OOP)

06 DESIGN & UX

Figma Wireframing Prototyping Design System
 Design Tokens Component Libraries Storybook
 UI/UX Design UI Implementation Responsive Design
 Mobile First Accessibility (WCAG) Usability User Flows
 Onboarding Flows Information Architecture UX Writing
 Microcopy Design Reviews

08 DEVOPS, CLOUD & INFRASTRUCTURE

Docker Docker Compose CI/CD Continuous Integration
 GitHub Actions GitLab CI Azure Pipelines Jenkins
 Nginx Sentry Monitoring Observability Linux
 Ubuntu Self-Hosting Microsoft Azure Azure Functions
 Azure App Service Azure Blob Storage Azure SQL
 Google Cloud Platform (GCP) Amazon Web Services (AWS)
 Hetzner IONOS Digital Ocean Vercel Supabase
 Firebase Cloud Functions Cloud Storage
 Firebase Cloud Messaging (FCM) Firebase Hosting DevOps
 DevSecOps

Skills & Expertise

Continued — Methodology, Compliance, Tooling & Domains

10 ARCHITECTURE & PRINCIPLES

Software Architecture Clean Architecture Clean Code
 SOLID Design Patterns Domain-Driven Design
 Separation of Concerns Dependency Injection
 Microservices Event-Driven Architecture Monorepo Nx
 Turborepo Scalable Architecture System Integration
 Platform Migration Middleware API-First
 Resilience Patterns Type Safety MVP Development

12 BUSINESS & STRATEGY

IT Strategy Digital Strategy Digital Transformation
 Process Digitization Product Ownership Entrepreneurship

14 TESTING & QA

Jest Vitest Cypress Playwright React Testing Library
 Unit Testing Integration Testing End-to-End Testing
 Postman API Testing Test Automation

16 DOMAINS & INDUSTRIES

LegalTech GovTech Law Firm Software Notary Services
 Client Portal Document Management HealthTech
 Medical Technology Maintenance Management ERP
 Inventory Management Warehouse Management Invoicing
 Order Management CRM Accounting
 Master Data Management Multi-Tenancy Stripe SEPA
 Payment Processing Logistics Route Planning
 PDF Generation Digital Archiving EdTech E-Learning
 Learning Management System (LMS) IoT Device Pairing
 Consumer Electronics Social Media SaaS
 Platform Development

11 METHODOLOGY & COLLABORATION

Agile Methods Scrum Kanban Scrumban
 Code Reviews Test-Driven Development (TDD)
 Software Development Project Management IT Consulting
 Requirements Engineering Remote Work Public Speaking

13 DATA PRIVACY & COMPLIANCE

GDPR Privacy by Design Data Minimization
 Data Processing Agreement (DPA) EU AI Act AI Governance
 GoBD Audit-Proof Record-Keeping EU Hosting
 Digital Sovereignty German Accessibility Act (BFSG)
 European Accessibility Act (EAA)

15 TOOLING

Git GitHub GitLab Husky Visual Studio Code
 Visual Studio Jira Linear Azure DevOps Bluedot
 HubSpot Cal.com Slack Google Workspace (Meet, APIs)
 Amplitude Product Analytics

17 ROLES & PROFILE

Senior Full-Stack Engineer AI Engineer Software Engineer
 Freelance Developer Interim CTO Technical Consultant
 Certified Application Developer (Fachinformatiker)
 German (Native) English B2 Munich DACH

Project Overview

All projects at a glance — with direct links to each detail page

01	BESCHEIDKLAR · Own Product Founder & CTO	LegalTech / GovTech	04/2025 – present ~15 months
02	Manifest OS · Manifest Law Full-Stack (AI features)	LegalTech / Immigration	12/2025 – 04/2026 ~5 months
03	RateUp · Krutzfeld Tech GmbH Sole Developer · Full-Stack	Social Media / Mobile App	11/2025 – 12/2025 ~2 months
04	AITOI · AITOI Frontend Engineer	IoT / Consumer Electronics	08/2025 – 09/2025 ~2 months
05	DiNo · LeXtorByte UG Frontend	LegalTech / Notary	01/2025 – 05/2025 ~5 months
06	AccountingOS · HD Autoservice Sole Developer	Retail & Automotive (Inventory Mgmt.)	01/2024 – 12/2024 ~12 months
07	Ladetrans · CDH Stephanus Sole Developer	Logistics & Transport	02/2021 – 05/2023 ~2 years 4 months
08	XU Navigator · XU Group Full-Stack	EdTech / E-Learning	05/2020 – 02/2023 ~2 years 10 months
09	Medical Device Management System · Hospital Group (anonymized) Full-Stack	HealthTech / Medical Devices	01/2019 – 05/2020 ~1 year 5 months

LEGALTECH / GOVTECH

04/2025 – PRESENT · 15 MONTHS

01

BESCHIEDKLAR — AI-Powered LegalTech SaaS PlatformRole: Founder & CTO Client: Own Product

Team: Solo (sole developer) · own product (side project)

Location: Lower Saxony, Germany · Remote

Project Language: German

Website: bescheidklar.de

Methodology: Self-directed · direct product ownership

RESPONSIBILITIES

- Designed and built a two-sided **LegalTech** SaaS platform with a modular, service-oriented architecture — combining AI-powered document analysis (**Azure OpenAI**) with a regional licensing model for law firms.
- Designed and implemented a multi-stage **AI pre-screening pipeline** for official notices and terminations (in progress) (extraction → deadline check → plausibility and success assessment) with structured case preparation.
- Automated, **GDPR-compliant** handoff of qualified leads (incl. uploaded/photographed documents, calculations, and AI results) directly into the system of matching specialists (lawyers, law clinics) — including contact details, producing an immediately actionable case lead.
- Built an AI-powered advice section with **RAG-based** content generation — articles and recommended actions for affected individuals (e.g. after termination or a rejection notice).
- Covered multiple legal domains (legal, social, and labor law) via an extensible, multilingual, low-barrier web interface.
- Full ownership of architecture, tech stack, EU-infrastructure hosting, and a secure-by-design compliance concept (**EU AI Act**, GDPR, German Legal Services Act).
- Built type-safe end-to-end data flows (**TypeScript**, **DTOs**, **Zod**) with automated **CI/CD** delivery (**GitHub Actions**, **Docker**), monitoring/error tracking (**Sentry**), and test coverage (**Vitest**, **Playwright**).

RESULTS

- Built the **MVP** independently — architecture, platform, and operations on EU infrastructure, all from a single hand.
- Compliance built into the product design from day one instead of bolted on later: **EU AI Act**, GDPR, and the limits of the German Legal Services Act (RDG) planned in from the start.
- Addressable market: 167,547 admitted lawyers in Germany (German Federal Bar, 1 Jan 2026); scalable model with minimal fixed costs.
- Target: up to 10,000 automated pre-screenings per month via AI-powered initial assessment.
- Target: up to **80%** less manual case intake through AI intake and automated lawyer routing.
- Target: predictable inflow of pre-qualified cases for partner law firms instead of costly manual initial assessment.

TECHNOLOGIES

TypeScript Python Next.js React FastAPI LLM Integration OpenAI API Azure OpenAI RAG (Retrieval Augmented Generation)
 Prompt Engineering Embeddings pgvector OCR Document Analysis LangChain PostgreSQL Redis Tailwind CSS shadcn/ui
 Stripe JWT OAuth 2.0 Docker GitHub Actions CI/CD Hetzner Linux Ubuntu Slack Husky Microservices
 Clean Architecture SaaS EU AI Act GDPR Privacy by Design Product Ownership

LEGALTECH / IMMIGRATION

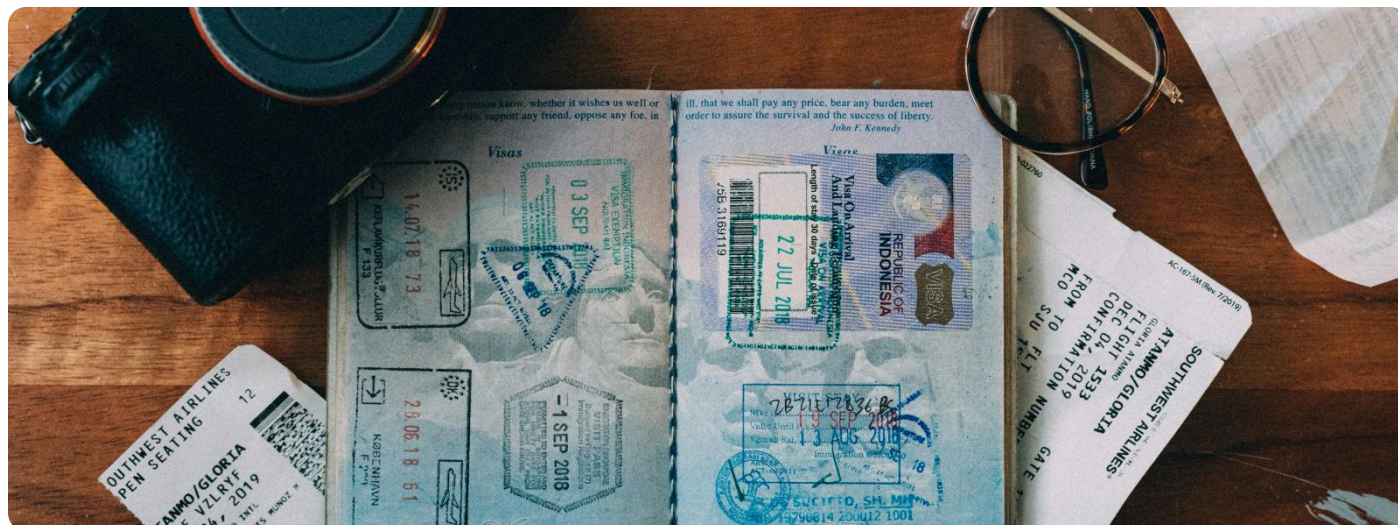
12/2025 – 04/2026 · 5 MONTHS

02

Manifest OS — AI-Powered Immigration Platform

Role: Full-Stack Engineer (AI Feature Integration) Client: [Manifest Law](#)

➤ Recommendation



Team: ~20 engineers in 4 teams (Platform, AI, Fullstack) · cross-functional collaboration with product, design, and tech leads

Location: New York, USA · Remote

Project Language: English

Website: manifestlaw.com

Methodology: Scrum · autonomous way of working

RESPONSIBILITIES

- Independently implemented production AI features: **event-driven** enrichment of attorney intake-call leads — meeting recordings are transcribed via **Bluedot** and processed resiliently (retry, idempotency) through a **webhook**, a **BullMQ** queue, and the OpenAI integration; structured results flow back automatically into the core systems.
- AI-powered evidence processing: uploaded case documents (e.g. ID scans, birth certificates, proofs) are automatically analyzed, classified, and consistently named.
- Contributed to consolidating four separate portals (B2B, B2C, Lawyer, Ops) as part of a four-engineer team and one API into a unified "**B2X**" portal; owned the full-stack migration of the B2B and Ops areas, including merging the data model and UI.
- Introduced app-wide event tracking (**Amplitude**) to analyze user behavior — identifying weak points and friction as a data foundation for product and data analytics.
- Built resilient, **event-driven** backend services (**NestJS**, **BullMQ**) with retry, idempotency, and error-handling strategies for reliable background processing.
- Delivered within a monorepo with clearly scoped feature modules to enable a fast future split into **microservices**, with type-safe interfaces (**TypeScript**, **DTOs**) and automated tests (**Jest**, **Playwright**) in the **CI/CD** pipeline.
- Worked closely in code reviews and used AI-powered tools (**Claude Code**, **Cursor AI**, **Code Rabbit**) to safeguard quality and development speed.

RESULTS

- Part of a roughly 20-person engineering team on an established platform (company valuation in the range of ~\$**60M USD**).
- Platform with over **3,000 customers**, 150+ corporate programs, and over 100 attorneys (team result).
- Up to **15%** higher approval rates versus the USCIS average (team result).
- AI features in production: intake-call leads and case documents are enriched, classified, and named without manual rework.
- Established app-wide event tracking as a shared data foundation for product and data analytics.
- Refactoring initiatives in the migrated B2B and Ops areas: simplified existing structures and improved long-term maintainability.

TECHNOLOGIES

TypeScript Python Next.js React NestJS Node.js Fastify Express.js REST API Webhooks DTOs BullMQ RabbitMQ

LLM Integration OpenAI API Anthropic Claude API Model Context Protocol (MCP) Bluedot HubSpot Cal.com Amplitude

PostgreSQL Redis MikroORM TanStack Query better-auth JWT Docker Linux Ubuntu Google Cloud Platform (GCP)

SOCIAL MEDIA / MOBILE APP

11/2025 – 12/2025 · 2 MONTHS

03

RateUp — Rating App for Web, iOS and AndroidRole: Sole Developer · Full-Stack Client: [Krutzfeld Tech GmbH](#)

Team: Solo (sole developer) · client project

Location: Zurich, Switzerland · Remote

Project Language: German

Website: krutzfeld.tech

Methodology: Self-directed

RESPONSIBILITIES

- Designed and built a social-media rating app (feed, ratings, profiles) from a single **Ionic/Angular** codebase — as an installable PWA on the web and as native builds for iOS and Android (**Capacitor**).
- Registration, login, and onboarding incl. profile creation via **Firestore Authentication**, secured with **Firestore Security Rules**.
- Built the social feed with filters, user posts, and media upload (**Cloud Firestore**, **Cloud Storage**).
- Server-side logic in **Cloud Functions**, with delivery via **Firestore** Hosting and a **CI/CD** pipeline (**GitHub Actions**) for all three platforms.
- Built a maintainable, handover-ready codebase: clearly layered structure, reusable components, type-safe models, unit tests, and consistent clean-code style (**SOLID**).

RESULTS

- Delivered a finished product to the client — running as a PWA on the web and as a native app on iOS and Android from a single shared codebase.
- Fully built on **Firestore**: authentication, data storage, media storage, server logic, and delivery.
- Handover-ready codebase: consistent structure, reusable components, unit tests, and full type safety — the client team can continue without an onboarding hurdle.
- From first line of code to handover in around two months.

TECHNOLOGIES

TypeScript Angular Ionic Capacitor RxJS Cross-Platform Development iOS Android Progressive Web App (PWA)
 Responsive Design UI Implementation User Flows Firestore Firestore Authentication (Auth) Firestore Security Rules Cloud Firestore
 Cloud Storage Cloud Functions Firestore Cloud Messaging (FCM) Firestore Hosting Authentication Onboarding Flows Social Media
 Type Safety Unit Testing Clean Code SOLID Separation of Concerns Clean Architecture Git GitHub GitHub Actions CI/CD
 Linux Trello

IOT / CONSUMER ELECTRONICS

08/2025 – 09/2025 · 2 MONTHS

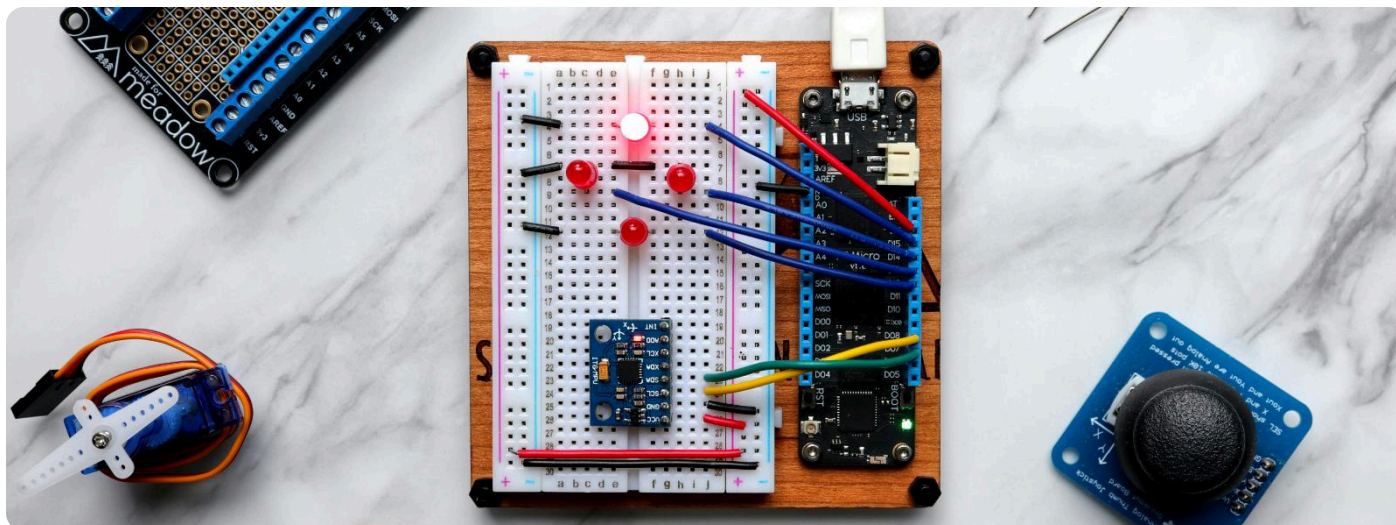
04

AITOI — Interactive IoT Toy (MVP)

Role: Frontend Engineer

Client: AITOI

➤ Recommendation



Team: Small cross-disciplinary team incl. hardware development

Location: Frankfurt, Germany · Remote

Project Language: German & English

Visibility: internal system (code not public)

Methodology: Self-directed · English-language coordination with hardware team

RESPONSIBILITIES

- Designed and built the IoT Wi-Fi onboarding for a connected AI toy via QR-code scan — incl. a guided setup flow, device pairing, and robust error handling.
- Implemented first-time device setup as well as later reconfiguration of settings via a platform-independent **Progressive Web App** (PWA).
- PWA dashboard for parents: track, filter, and analyze children's play sessions.
- Integrated **Supabase** for authentication, data storage, and real-time sync between device and app (incl. **row level security**).
- Built a performant, platform-independent PWA architecture (**Next.js**) with a responsive UI.
- Defined and implemented type-safe data models and API interfaces (**TypeScript**, **DTOs**, **Zod**).
- Coordinated closely, in English, with the hardware team to define device-app communication and the setup protocols.
- Advised and guided the client from initial idea to an actionable **MVP** concept.

RESULTS

- Fully functional **MVP** frontend delivered in ~2–3 weeks of actual working time — taken independently from concept through client coordination to the demo.
- Platform-independent PWA with real-time sync between the IoT device and the app.
- Scalable, documented architecture as a **solid** technical foundation for the product launch.
- Trusted, cross-functional collaboration with the hardware team throughout the project — despite the tight timeline, no friction at the device-app interface.

TECHNOLOGIES

TypeScript Next.js React Progressive Web App (PWA) Responsive Design Design System React Hook Form DTOs
 Device Pairing Supabase Supabase Realtime Row Level Security PostgreSQL Tailwind CSS shadcn/ui JWT Authentication
 Docker Linux Ubuntu CI/CD GitHub Actions Vitest Cursor AI Model Context Protocol (MCP) MVP Development
 Clean Architecture

LEGALTECH / NOTARY

01/2025 – 05/2025 · 5 MONTHS

05

DiNo — Digital Notary

Role: Frontend Engineer

Client: [LeXtorByte UG](#)

📌 Recommendation



Team: 2–4 people

Location: Oranienburg, Germany · Remote

Project Language: German

Website: digitales-notariat.de

Methodology: Scrumban

RESPONSIBILITIES

- Implemented frontend features to digitize notary processes with a focus on clear, traceable user flows — incl. client portal and notary portal.
- Built digital case management: structured data capture, digital case files, and seamless internal filing to speed up case processing.
- Contributed to extending a client portal for status tracking, secure document delivery, and privacy-compliant communication.
- Built reusable components and unified form validation (**React Hook Form**) to speed up feature development across the team.
- Coordinated closely with design and backend in a **Scrumban** rhythm to quickly translate new requirements into usable interfaces.
- Took initiative in the codebase: refactored existing code and fixed legacy bugs — geared toward durable, long-term solutions.

RESULTS

- Per the provider, up to **70%** fewer client phone inquiries.
- Per the provider, invoices are paid up to **30 days** sooner.
- Noticeably reduced administrative overhead and greater transparency through consistently digital, traceable processes.
- Consistent, modern UI across the client and notary portals, strengthening both client trust and notary-office efficiency.

TECHNOLOGIES

TypeScript React Python Flask REST API OpenAPI DTOs MariaDB MySQL Redux React Router React Hook Form
 Material UI (MUI) Single Page Application (SPA) User Flows Client Portal Digital Archiving JWT OAuth 2.0
 Role-Based Access Control (RBAC) GDPR Docker Nginx Linux Ubuntu GitLab Slack Clean Architecture

RETAIL & AUTOMOTIVE (INVENTORY MGMT.)

01/2024 – 12/2024 · 12 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

06

AccountingOS — GoBD-Compliant Accounting & Inventory System**Role:** Sole Developer · Full-Stack & AI **Client:** HD Autoservice

📌 Recommendation

**Team:** Solo (sole developer) · client project**Location:** Salzgitter, Germany · Remote**Project Language:** German**Visibility:** internal system (code not public)**Methodology:** Self-directed**RESPONSIBILITIES**

- Designed and built a feature-rich, **GoBD-compliant accounting** and **inventory management** system on top of a cleanly layered architecture (**Clean Architecture**).
- Designed and implemented a resilient data-sync service ("pipe service") that processes supplier catalogs (auto parts) with up to **20M** CSV rows over a TLS-encrypted connection — with queuing, caching, retry logic, validation, deduplication, and data cleansing.
- Built the **accounting** module: invoices, credit notes, internal orders, PDF export, customer management, **inventory management**, order import with automatic order conversion, and **GoBD** record-locking.
- Coupled **inventory management** with invoice management: automatic stock deduction on invoice line items and notification when falling below minimum stock.
- Recursive resolution of foreign-key dependencies in input dialogs for consistent master-data maintenance.
- Customer and master-data management with **role-based access control** (admin, staff, tax advisor) and authentication via **better-auth**.
- Advised the client on technical approaches, architecture, and process optimization.
- Built type-safe **GraphQL** and REST interfaces (**NestJS**) with validation, error handling, and consistent data modeling (**MikroORM**, **DTOs**).
- Ongoing support (side project): maintenance, security updates, and further development.

RESULTS

- Per the client, saves up to **40 hours** of admin work per month (up to **480 hours/year**) — about €19,200 a year at an assumed labour cost of €40 per hour.
- Cuts time spent on routine processes like invoicing and **inventory management** by up to **50%**.
- Consistently digital, **GoBD-compliant** workflows with direct coupling of inventory and **accounting**.

TECHNOLOGIES

TypeScript Node.js NestJS Next.js React Fastify REST API GraphQL DTOs Type Safety Data Synchronization
 CSV Processing Message Queues Caching MariaDB PostgreSQL Redis MikroORM TanStack Query Tailwind CSS
 Inventory Management Warehouse Management Accounting ERP GoBD better-auth JWT Role-Based Access Control (RBAC)
 Docker Self-Hosting Linux Ubuntu CI/CD GitHub GitHub Actions Husky Claude Code Cursor AI
 Model Context Protocol (MCP) Microservices Clean Architecture

LOGISTICS & TRANSPORT

02/2021 – 05/2023 · 2 YEARS 4 MONTHS · ONGOING MAINTENANCE (SIDE PROJECT)

07

Ladetrans — Truck Route Management for a Humanitarian Aid Organisation

Role: **Sole Developer** · Full-Stack & AI Client: **CDH Stephanus**



Team: Solo (sole developer) · client project

Location: Bremen, Germany · Remote

Project Language: German

Website: cdh-stephanus.org

Methodology: Self-directed

RESPONSIBILITIES

- Load lists per tour: goods to be shipped, recipient addresses, and drivers assembled in a single step.
- Each tour as one continuous process — create, edit, complete, and archive.
- Automated generation of tour documents from the captured data, incl. for border checks.
- Audit-proof archiving of completed tours, including the proof-of-delivery files uploaded at tour end.
- Sole responsibility for backend architecture (**NestJS**, **Clean Architecture**), **role-based access control (better-auth)**, and the self-hosted deployment (**Docker**, **Nginx**) incl. ongoing maintenance (side project).

RESULTS

- Per the client, saves up to **1,000 hours** per employee per year through automated document processes — about €40,000 annually at an assumed labour cost of €40 per hour.
- Per the client, up to **30%** higher **logistics** efficiency through the end-to-end management of shipping tours.
- Significantly reduced transcription errors and costly rework through consistently digital processes.

TECHNOLOGIES

TypeScript Node.js NestJS Next.js React Fastify REST API GraphQL PostgreSQL Redis MikroORM React Router
 Tailwind CSS Single Page Application (SPA) Logistics Route Planning PDF Generation DOCX Generation Digital Archiving
 better-auth JWT Authentication Role-Based Access Control (RBAC) Docker Self-Hosting Linux Ubuntu CI/CD
 GitHub Actions Husky Claude Code Cursor AI Model Context Protocol (MCP) Slack Nginx Microservices Clean Architecture

EDTECH / E-LEARNING

05/2020 – 02/2023 · 2 YEARS 10 MONTHS

08

XU Navigator — Enterprise Learning Platform

Role: Full-Stack Engineer

Client: **XU Group**

➤ Recommendation



Team: 5–10 people

Location: Bremen, Germany · Remote

Project Language: German & English

Website: xu.de

Methodology: Scrum

RESPONSIBILITIES

- Co-developed "XU Navigator", a scalable microservice learning platform for enterprise employees — for clients including Daimler, VW, and ThyssenKrupp.
- Built dynamically, tenant-specific course content (videos, podcasts, multiple-choice quizzes, articles) across clearly scoped services.
- Implemented learning-progress tracking incl. automatic certificate generation as well as an expert and community forum.
- Full-stack development across all platform modules within an agile **Scrum** team — from course management to the certificate engine, in both frontend and backend.
- Built scalable backend services (**NestJS**, **GraphQL**) with clearly scoped domains in a microservice architecture.
- Built responsive **Angular** interfaces (**Angular Material**, **NgRx**) focused on reusability and maintainability.
- Deployed via Azure (**Azure DevOps** Pipelines, **Docker**) with automated builds and deployments.

RESULTS

- Scalable microservice platform in production use at major enterprises (Daimler, VW, ThyssenKrupp, and others).
- High degree of customization for tailored, company-specific training programs.
- Automated certificate issuance and progress analytics to boost completion rates.
- Close, cross-functional collaboration with product and design over two and a half years — from initial rollout to production use at multiple enterprise clients.

TECHNOLOGIES

TypeScript Angular Node.js NestJS Express.js REST API GraphQL MongoDB Mongoose NgRx RxJS Angular Material
 SCSS Figma Microsoft Azure Azure DevOps Azure Pipelines Azure App Service Azure Blob Storage Docker CI/CD JWT
 OAuth 2.0 Single Sign-On (SSO) Learning Management System (LMS) Video Streaming Microservices Clean Architecture

HEALTHTECH / MEDICAL DEVICES

01/2019 – 05/2020 · 1 YEAR 5 MONTHS

09

Medical Device Management System

Role: Full-Stack Engineer **Client:** Hospital Group (anonymized)**Team:** 2–4 people**Location:** Germany · On-site / Hybrid**Project Language:** German**Visibility:** HealthTech project for a hospital group (anonymized)**Methodology:** Scrumban

RESPONSIBILITIES

- Co-developed all areas of software to manage medical devices across multiple hospitals — a tool for technicians to simplify maintenance and repair.
- Built device and master-data management (hospital, location, device specifications) on top of a cleanly layered architecture (**Clean Architecture, Dependency Injection**).
- Developed a detailed maintenance history (timing and type of work performed) plus device-linked document management.
- Implemented fault reporting by nursing staff for fast handoff to technicians.
- Built multi-tenant filtering by hospital plus integration of Microsoft sign-in via **O365 Identity**.
- Built a type-safe **REST API (ASP.NET Core Web API, Swagger/OpenAPI)** with clean layering and **dependency injection**.
- Data modeling and access with **Entity Framework Core** (code-first migrations) on **MSSQL**.
- Automated builds via **Azure DevOps** Pipelines.

RESULTS

- Delivered complete, multi-tenant device management with maintenance history and document management.
- Structured technical foundation for more efficient maintenance and repair processes.
- Contributed for over a year as part of a small, closely aligned team on a safety-critical application in a hospital setting — with a focus on rigor, traceability, and stable delivery.

TECHNOLOGIES

C# .NET Core ASP.NET Core ASP.NET Core Web API OpenAPI Swagger TypeScript React Entity Framework Core
 Data Modeling Microsoft SQL Server (MSSQL) Fluent UI Redux React Router Axios Single Page Application (SPA)
 Dependency Injection JWT O365 Identity Role-Based Access Control (RBAC) Multi-Tenancy Master Data Management
 Document Management Microsoft Azure Azure SQL Azure DevOps Docker CI/CD Clean Architecture